Neuron mechanisms I: how $d2$-dominant neurons compute pair parities

`wd3_k0_N2048`, step 500k — post-nonlinearity units, uniform evaluation data

Setup and mechanism

For each neuron we analyze $a = \phi(z)$, with $\phi = \text{relu}(\tanh)$ and $z = (Vh)_n$, evaluated on all 2^{16} uniform inputs. We call a neuron $d2$-*dominant* when its largest coefficient among the tracked tree modes is a degree-2 pair parity $\chi_{x_i x_{i+1}}$. Such neurons make up most of the active population in every block (1528/1330/1061/1036 active per block).

The exemplar (block 2, neuron 1779, pair $d2_4 = (x_8, x_9)$) makes the mechanism particularly clear (Fig. 1). Its pre-activation is almost exactly a *balanced difference of the two singleton inputs*, $z \approx \gamma(x_8 - x_9)$ with $\gamma = 0.173$ and residual std 0.004. Therefore $a = \phi(z)$ is positive only at the single corner $(x_8, x_9) = (+, -)$, where its value is $\tanh(2\gamma) \approx 0.35$.

A detector for one corner of the two-bit hypercube has the Walsh decomposition $a = \frac{t}{4}(1 + x_8 - x_9 - x_8 x_9)$. Thus it contains **four Walsh coefficients of equal magnitude**: a constant term, the two singleton modes, and the pair parity. Since the constant term does not contribute to activation variance, the variance is split into exact thirds over $\{x_8, x_9, x_8 x_9\}$ (measured 0.330/0.330/0.327).

The target tree does not include degree-1 modes, so only the $x_8 x_9$ component is counted as tracked tree power. This is why the tree coverage is 1/3 *exactly by the structure of the corner detector*: the remaining two thirds are the two singleton components x_8 and x_9, each with essentially the same coefficient magnitude as the pair itself.

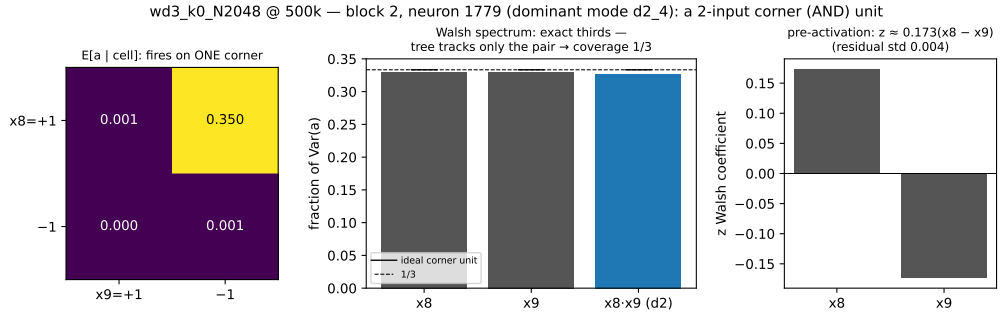


Figure 1: The $d2$ corner unit (block 2, #1779). Left: it fires on one corner of (x_8, x_9). Middle: measured Walsh spectrum (bars) vs. the parameter-free ideal corner unit (ticks) — exact thirds; only the pair (blue) is tree-tracked. Right: the pre-activation is a balanced singleton difference.

Population: every active $d2$ neuron is a corner unit

The exemplar is not an isolated case. Rank the $d2$-dominant neurons in each block by their pair power a^2. Among all neurons before the dead/noise tail, the fraction whose pre-activation is a balanced two-singleton combination is **1.0 in every block**. Here "balanced" means that the two relevant singletons account for at least 90% of $\text{Var}\, z$ and their coefficient magnitudes satisfy $|\alpha|/|\beta| \geq 0.8$. For the top 256 neurons in each block, the median fraction of pre-activation variance coming from the two relevant singletons is 0.993–0.999 (Fig. 2).

All four possible corner-detector sign patterns are represented. Roughly half of the neurons are difference type, $z \propto x_i - x_{i+1}$, which fires on a corner with pair parity -1 and therefore has a negative pair coefficient. The other half are sum type, $z \propto x_i + x_{i+1}$, which fires on a corner with pair parity $+1$ and therefore has a positive pair coefficient.

This balanced population is exactly what allows the network to reconstruct the full ± 1 pair parity downstream. Individual neurons carry the desired pair mode together with singleton by-products, but an appropriate weighted sum across many corner units can reinforce the pair component while cancelling the singleton components.

Consistent with the $d4/d8$ reports, these pair features are *end products* for the supervised $d2$ targets rather than intermediate building blocks for the higher-degree parities. The deeper parity modes are generated by separate raw-bit majority-gate mechanisms rather than by composing these $d2$ corner units.

Artifacts: `gdrive:parity/analysis_output/neuron_mechanisms/wd3_k0_N2048/step_00500000/` (`per_neuron.csv`, figures, this report; companion $d4/d8/d16$ reports in the same folder).

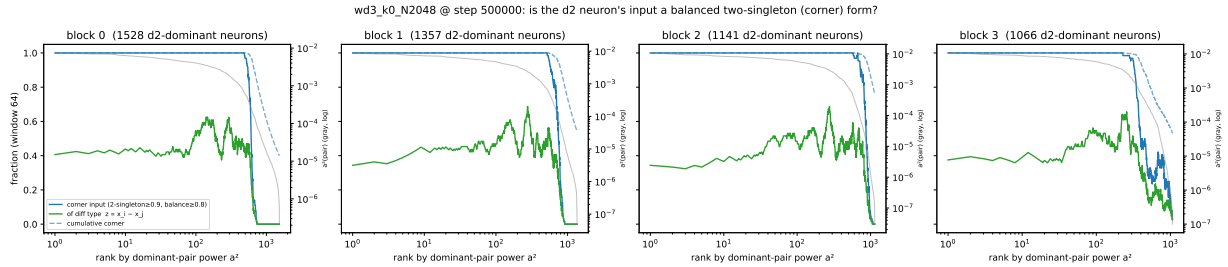


Figure 2: Fraction of $d2$-dominant neurons with corner input vs. rank by pair power (blue, pinned at 1.0 until the noise tail; gray: a^2 of the pair); green: the strict difference type $z \propto x_i - x_{i+1}$ ($\approx 1/2$).